DSGE Model Estimation via VAR IRF Matching: The Short and Long of It *

Lorenzo Del Col[ξ]

October, 2026

Abstract

Following De Graeve and Westermark (2025) and Baumeister (2025), I increase the lag length of the VAR in Altig et al. (2005) to re-estimate their DSGE. Moving from $p = 4$ to $p = 12$ lags, I find that both the VAR and DSGE impulse responses, and the implied structural parameters, change substantially. The Calvo price parameters that distinguish the observationally equivalent firm-specific and homogeneous capital versions of the model converge to one another, from 0.34 against 0.82 at $p = 4$ to 0.89 against 0.90 at $p = 12$. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters, equal to that of the homogeneous capital model, re-opening the micro–macro discrepancy in price stickiness that firm-specific capital was introduced to resolve.

Keywords: Truncation Bias, Long-Lag VARs, DSGE Estimation, IRF Minimum Distance , Nominal Rigidities
JEL Codes: C32, C52, E32

*...

[ξ]University of Padova. Contact: lorenzo.delcol@studenti.unipd.it

Contents

1 Introduction **1**

1.1 Motivation 1

1.2 Contribution and Results 2

1.3 Related Literature 3

1.4 Outline 3

2 The Theoretical Model **3**

2.1 Inflation inertia and the micro–macro conflict 3

2.2 Homogeneous and firm-specific capital 3

3 Truncation Bias and the Case for Long-Lag VARs **4**

3.1 A simple illustration 4

4 Empirical Results **5**

4.1 DSGE structural parameters 5

4.2 Implied duration and the collapse of the firm-specific mechanism 6

5 Conclusion **6**

A Data **9**

B The ACEL Model: Full Derivation **9**

C Additional Residual Analysis **9**

D Monte Carlo Inference on the Calvo Parameters **9**

E Robustness **9**

F Replication Files and Code **9**

1 Introduction

1.1 Motivation

An important goal of monetary business cycle research is to explain the empirical evidence on the impact of economic shocks on macroeconomic variables. Because the systematic component of policy is endogenous, identifying the causal effect of monetary policy requires isolating the non-systematic component, the monetary policy shock, even though it is small and infrequent (Ravenna, 2007; Aruoba and Drechsel, 2024).

A large class of estimated DSGE models is taken to the data through this channel. Impulse Response Function Matching Estimation (IRFME) is a limited-information strategy that chooses structural parameters to minimise the distance between the impulse responses implied by the model and those estimated from an identified VAR (Christiano et al., 2005; Altig et al., 2011; Rossi et al., 2012). The strategy is attractive precisely because it is partial: the researcher takes a stand on the identification of a subset of shocks, and remains agnostic about the remaining dimensions of the data.

That attractiveness rests entirely on one maintained assumption: that the VAR delivers *consistent* estimates of the moments being matched. If the reduced-form VAR adequately captures the data generating process, impulse responses are optimal at all horizons; if the VAR is misspecified, the specification error propagates to the structural statistics and compounds at each horizon (Ramey, 2016; Canova and Fosso, 2025) Ravenna (2007) "The bias introduced by truncation can lead to bias in the identification of the structural shock." Altig et al. (2011) say as much themselves: "the quality of our estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks".[1]

Since a VAR(∞) cannot be estimated on a finite sample, researchers truncate. The resulting VAR(p) is misspecified, and increasingly so the smaller is p (De Graeve and Westermark, 2025; Barnichon et al., 2026; Ravenna, 2007). Ravenna (2007) "the problem facing economistis is whether the number of lags p to be included is reasonable given the lenght of economics time series over which VARs are estimated. When estimating a VAR consistent with DSGE business cycle models it is standard to assume that including few lags is sufficient to provide a reasonable approximation to the exact infinite order VAR representation! De Graeve and Westermark (2025) show that the conventional objection to longer lags, that the additional parametrisation makes confidence intervals prohibitively wide, is incomplete, because reducing misspecification reduces not only bias but also variance. Their conclusion is addressed explicitly to applied researchers: "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice".

[1] Altig et al. (2011, footnote 10, p. 3).

1.2 Contribution and Results

This paper asks what lag truncation does to *structural* estimates, not merely to impulse responses. I replicate Altig et al. (2011) — henceforth ACEL — and re-estimate their DSGE by IRF matching against VARs of increasing lag order.

The contribution is twofold. First, relative to De Graeve and Westermark (2025), I move the question from the reduced form to the structural parameters. Second, and more specifically, I do so in a setting their design deliberately excludes. De Graeve and Westermark (2025) state that "all our experiments are based on two-shock models and two-variable VARs", and that where a candidate model does not fit that framework — Smets and Wouters — "we consider the model with only monetary policy and preference shocks".[2] Their data generating processes are square and fundamental by construction, so that truncation bias is isolated from singularity and invertibility. ACEL's model is not: it confronts three structural shocks with a ten-variable VAR. [**TBD:** Verificare che nessun altro abbia gia' fatto questo esercizio su un modello stocasticamente singolare. Se confermato, e' il claim centrale.]

My main results are the following. *First*, the impulse responses to a monetary policy shock retain their sign and broad shape across lag orders, but the humps peak substantially later as p increases, and the point estimates remain inside the VAR(4) confidence bands. *Second*, the responses to the two technology shocks — identified with long-run rather than short-run restrictions — do not: several fall outside the baseline confidence set. This asymmetry is what Ravenna (2007) predicts, since long-run restrictions require the sum of the entire lag polynomial and truncation error accumulates in exactly that object. *Third*, the structural estimates move. The Phillips curve slope falls from 0.0402 to 0.0129 and becomes precisely estimated only at $p = 12$; the cost of varying capital utilisation collapses from 1.99 to 0.004; the investment adjustment cost roughly doubles. *Fourth*, and as a consequence, the implied Calvo price parameters of the firm-specific and homogeneous capital models converge, from 0.34 against 0.82 to 0.89 against 0.90. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters.

The last result overturns ACEL's central conclusion. Their claim is that only the firm-specific capital model reconciles macro evidence on inflation inertia with micro evidence on the frequency of price adjustment. Under a richer lag structure that reconciliation disappears. The channel is transparent and internal to their own argument: ACEL note that "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[3] Long-lag VARs push the estimate of σ_a precisely into that region.

[2] De Graeve and Westermark (2025, p. 10). They also note that "invertibility is never a problem; all the models we consider are fundamental".

[3] Altig et al. (2011, p. 227).

1.3 Related Literature

Long-lag VARs.
Truncation bias and the DSGE–VAR mapping.
Impulse response matching and DSGE estimation.

1.4 Outline

Section 3 illustrates the truncation bias problem, separates its two channels following Ravenna (2007), and reviews the case for longer lags. Section 2 presents the ACEL model and the role of firm-specific capital in reconciling micro and macro evidence on price setting. Section **??** describes the econometric strategy: why a stochastically singular DSGE is estimated by limited-information methods, and what that strategy assumes. Section 4 presents the empirical results. Section 5 concludes.

2 The Theoretical Model

2.1 Inflation inertia and the micro–macro conflict

Altig et al. (2011) start from a tension in the New Keynesian literature. To account for inertia in inflation, standard models require firms to re-optimise prices infrequently — every six quarters, and in versions without indexation to past inflation even every two years. This sits uneasily with the microeconometric evidence, which finds price changes considerably more frequent. [**TBD:** Citare Bils and Klenow (2004), Golosov and Lucas (2007), Klenow and Kryvtsov (2008), che sono i riferimenti di ACEL. Valutare se aggiungere evidenza micro piu' recente. Da aggiungere a biblio.bib.]

Altig et al. (2005) propose a resolution. In their model inflation is inertial even though firms re-optimise on average every 1.5 quarters. Inertia arises not because firms change prices rarely, but because when they do they change them by a small amount: each firm's short-run marginal cost is increasing in its own output, which dampens the optimal price adjustment. The assumption delivering this is that capital is predetermined within the period and firm-specific, so that factors of production cannot be instantly reallocated across firms.

2.2 Homogeneous and firm-specific capital

ACEL conduct the analysis using two versions of the Christiano et al. (2005) model: one in which capital is homogeneous, as in standard formulations, and one in which it is firm-specific. Once log-linearised, the only difference between the two sets of equilibrium

conditions lies in the equation relating inflation to marginal cost,

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \tag{1}$$

where $\hat{s}_t$ is economy-wide average real marginal cost in units of the final good, and the slope γ maps into the structural parameters as

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi. \tag{2}$$

The information set Ω_t contains the current realisation of the technology shocks but not that of the monetary policy innovation.

The two models are parametrised in terms of γ rather than ξ_p. Given the remaining parameters they are then observationally equivalent with respect to aggregate prices and quantities, so that no stand need be taken on which version is being estimated. What distinguishes them is the value of ξ_p that a given γ implies.

3 Truncation Bias and the Case for Long-Lag VARs

VARs are subject to at least three sources of bias: small-sample bias, identification bias (Forni et al., 2016), and truncation bias (De Graeve and Westermark, 2025; Ravenna, 2007; Chari et al., 2008; Barnichon et al., 2026). This paper is concerned with the latter, and with the way it contaminates the second.

3.1 A simple illustration

Following De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{3}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{4}$$

delivers a biased point estimate unless the regressors are orthogonal, $\mathbb{E}(b_1) \neq \beta_1$, and an upward-biased residual variance, $s^2 > \sigma^2$. Both distortions matter here, and the second is the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{5}$$

with $B(L) = B_1L + \cdots + B_pL^p$ and $p \ll p^*$, where p^* denotes the true lag order. Truncation biases both the autoregressive coefficients $B(L)$ and the reduced-form covariance matrix Σ_u. Since impulse responses are a function of both,

$$\text{IRF} = f\big(B(L), \Sigma_u\big), \tag{6}$$

they inherit the bias.

4 Empirical Results

4.1 DSGE structural parameters

Table 1: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.8213	0.8958
	(0.0479)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.6448	0.8884
	(0.1503)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The Calvo price parameters $\xi_{p,hom}$ and $\xi_{p,fs}$ are not estimated directly: they are recovered numerically from the estimated structural parameters. Their standard errors are therefore obtained by Monte Carlo simulation with 1,000,000 draws from the joint distribution of the estimated structural parameters, solving for the implied Calvo parameter at each draw.

4.2 Implied duration and the collapse of the firm-specific mechanism

Table 2: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR ($p = 4$)**: Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

The Chi parameter

Table 3: Implied Strategic Complementarity Parameter χ Across Lag Lengths

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model (validity check, $\chi \equiv 1$)			
χ_{hom}	0.9989	1.0010	0.9977
	(0.0000)	(0.0000)	(0.0000)
Firm-Specific Capital Model			
χ_{fs}	0.0307	0.2030	0.8671
	(0.0596)	(0.1667)	(0.1698)

Note: χ is recovered by inverting $\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi$ at the estimated γ and implied ξ_p. $\chi_{hom} \equiv 1$ by construction (Proposition 1(i) in Altig et al., 2011); reported here only as a check that the numerical pipeline is consistent. Standard errors in parentheses are computed from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters' standard errors, applying the formula above draw-by-draw. The 90% Monte Carlo percentile interval for χ_{fs} is [0.0067, 0.0879], [0.0516, 0.5101], [0.3704, 0.9270] for $p = 4, 8, 12$ respectively; see the discussion of non-normality of the Calvo/chi Monte Carlo distributions in the text.

5 Conclusion

References

Altig, D., Christiano, L. J., Eichenbaum, M., and Lindé, J. (2005). Firm-specific capital, nominal rigidities and the business cycle. NBER Working Paper 11034, National Bureau of Economic Research. Working paper predecessor of the published version (Review of Economic Dynamics, 2011).

Altig, D., Christiano, L. J., Eichenbaum, M., and Lindé, J. (2011). Firm-specific capital, nominal rigidities and the business cycle. *Review of Economic Dynamics*, 14(2):225–247.

Aruoba, S. B. and Drechsel, T. (2024). Identifying monetary policy shocks: A natural language approach. Working paper, University of Maryland, NBER, and CEPR.

Barnichon, R., Mesters, G., Nakamura, E., and Steinsson, J. (2026). Humble vars. Working paper, University of California, Berkeley.

Baumeister, C. (2025). Discussion of "local projections or vars? a primer for macroeconomists". *NBER Working Paper Series*.

Canova, F. and Fosso, L. (2025). Low frequency movements and svar analyses. CAMP Working Paper 10/2025, Centre for Applied Macroeconomics and Commodity Prices (CAMP).

Chari, V. V., Kehoe, P. J., and McGrattan, E. R. (2008). Are structural vars with long-run restrictions useful in developing business cycle theory? *Journal of Monetary Economics*, 55(8):1337–1352.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1):1–45.

De Graeve, F. and Westermark, A. (2025). Long-lag vars. *Journal of Monetary Economics*, 156.

Forni, M., Gambetti, L., and Sala, L. (2016). Var information and the empirical validation of dsge models. RECent Working Paper 119, RECent - Center for Economic Research, University of Modena and Reggio Emilia.

Ramey, V. A. (2016). Macroeconomic shocks and their propagation. In *Handbook of Macroeconomics*, volume 2, pages 71–162. Elsevier.

Ravenna, F. (2007). Vector autoregressions and reduced form representations of dsge models. *Journal of Monetary Economics*, 54(7):2048–2064.

Rossi, B., Hall, A. R., Inoue, A., and Nason, J. M. (2012). Information criteria for impulse response function matching estimation of dsge models. *Journal of Econometrics*, 170(2):499–518.

A Data

[**TBD:** Tabella con le dieci variabili: definizione, fonte, trasformazione. Sul modello della Table A.1 di Lami.]

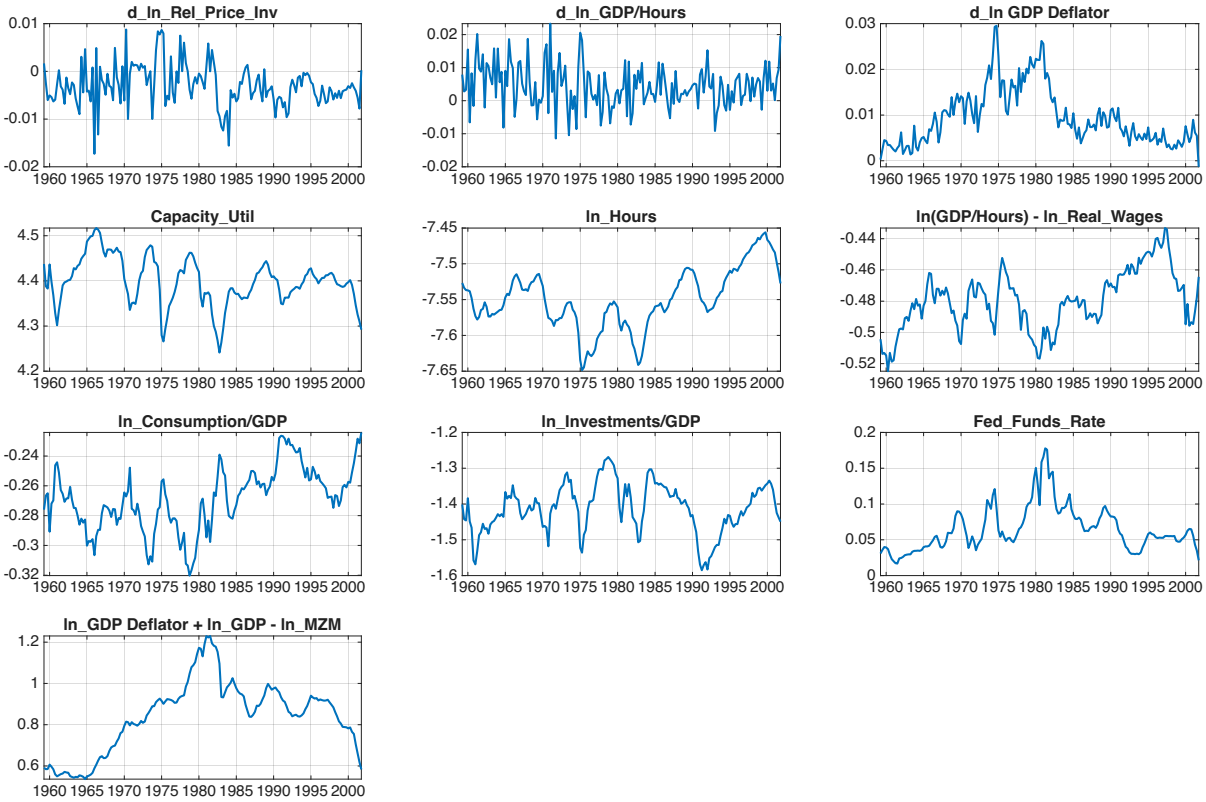


Figure 1: Raw historical series

B The ACEL Model: Full Derivation

C Additional Residual Analysis

D Monte Carlo Inference on the Calvo Parameters

E Robustness

F Replication Files and Code